Lab/Project Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
					Total obtained mark	
Comments						

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Student Name: Rashidul Hasan, Ahiatul Jannat, Tanjina Akter Akhi, Taspia Ahamed, MD. Badhon Safayet Lam

Student ID: 232-35-638, 232-35-538, 232-35-507, 232-35-710, 232-35-753

Batch: 41 **Section:** M

Course Code: SE323 **Course Name:** Software Engineering Web Application Lab

Course Teacher Name: Rantu Das

Designation: Lecturer, Department of Software Engineering

Submission Date: 17 / 04 / 2026

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DIU Lost & Found System

1. Project Overview

1.1. Introduction

The **DIU Lost & Found** system is a **web-based platform** developed to help students, faculty, and staff at **Daffodil International University (DIU)** efficiently manage and recover lost items on the university campus. This system aims to streamline the entire process, from reporting lost items to tracking down found items, ensuring a smooth and systematic recovery process.

1.2. Problem Statement

At DIU, students and staff often lose important items such as **ID cards, wallets, mobile phones, electronic devices**, and various personal belongings. Currently, there is no centralized system to track and recover these items efficiently. This leads to:

- Increased time spent searching for lost items by students and staff.
- Lack of communication between the people who find items and those who have lost them.
- Inconsistent and unstructured reporting of lost and found items, often done through informal channels like word-of-mouth or personal messaging.

1.3. Objective

The primary objective of this project is to provide an efficient, user-friendly, and reliable platform that offers:

- Easy reporting of lost and found items: Users can quickly post lost or found items with relevant details, including descriptions, images, and contact information.
- Browsing found items by category: A categorized list of found items will be available for users to browse, making it easier for them to locate their lost belongings.
- Effective communication between users and admins: The system will have a built-in messaging system for users to inquire about specific items and for admins to manage inquiries and claims.

2. System Design & Architecture

The DIU Lost & Found system is designed using modern web technologies to ensure a user-friendly interface and efficient backend. The architecture is built for scalability, efficiency, and security, supporting key features like posting items, tracking lost items, and facilitating communication between users and admins.

2.1. Frontend Architecture

The frontend is developed to be simple, intuitive, and responsive across devices. The primary technologies used are:

- **HTML & CSS:** These provide the structure and design of the website, ensuring it is visually appealing and works well on both desktop and mobile devices.
- **JavaScript:** Used to implement dynamic features such as form validation, search functionality, and filtering of items based on category, location, or status.

2.1.1. Frontend Workflow

- **Homepage:** The homepage displays available categories, featured items, and provides options for users to browse or post items.
- **Item Details:** Each item posted has its own page displaying all relevant information such as description, status, and the contact details of the person who found it.
- **Search and Filter:** Users can search for items by keywords or filter them by category, location, or date they were found.

2.2. Backend Architecture

The backend is built using PHP, which handles all server-side operations. It manages the interactions with the MySQL database and processes business logic for core functionalities.

- **PHP:** Handles HTTP requests and processes item submissions, approvals, and messaging between users and admins.
- **MySQL:** A relational database system used to store all information related to the lost and found process. It is chosen for its scalability and ease of use with PHP.

2.2.1. Backend Workflow

1. **User Requests:** When a user submits a request (like posting a found item or submitting an inquiry), PHP processes the request, validates the data, and stores it in the MySQL database.

2. **Admin Management:** Admins can access a secure dashboard to approve or delete found items, manage user inquiries, and view analytics regarding the status of items.

3.Database Design

The **MySQL database** is structured to store and manage different types of data efficiently. The core database tables include:

- **category_list**: Stores item categories (e.g., Electronics, ID Cards, Bags) to help organize found items.
    - **Fields**: category_id, category_name, category_description
- **item_list**: Stores detailed information about each found item.
    - **Fields**: item_id, category_id (foreign key), item_title, item_description, item_status (Pending, Claimed, Expired), item_location, item_contact_details, item_image
- **inquiry_list**: Stores inquiries made by users related to specific items.
    - **Fields**: inquiry_id, item_id (foreign key), user_name, user_message, inquiry_status
- **users**: Stores user data, which may include admin roles for system management.
    - **Fields**: user_id, user_name, user_role, user_email, user_password
- **system_info**: Stores system-wide settings and configurations like system time zone, date formats, and notification preferences.
    - **Fields**: system_id, system_config_key, system_config_value

3.1. Data Flow

1. **Item Submission**: Users can submit items, and these items are stored in the item_list table. Once approved by the admin, they are marked as Published.
2. **Admin Actions**: Admins can access and modify the item records, which might include approving pending items or marking them as Claimed or Expired.
3. **User Inquiries**: Inquiries about specific items are stored in the inquiry_list table, and admins can reply accordingly.

4.Functional Features

4.1. System Features

No.	Feature	Description
1	**Lost Item Listing**	Displays a list of lost items with details like description, status, and contact.
2	**Category Filtering**	Allows users to filter items by predefined categories like electronics, bags, etc.
3	**Item Details Page**	Dedicated page for each item with detailed information and contact details.
4	**Search Functionality**	Users can search for lost or found items by entering keywords.
5	**Contact Form**	Allows users to contact the admin or item poster for further communication.
6	**Admin Dashboard**	Admins can manage items, categories, and messages through the backend dashboard.
7	**Item Management (CRUD)**	Admins can create, read, update, or delete items posted by users.
8	**Category Management (CRUD)**	Admins can create, read, update, or delete categories to organize items.
9	**Item Image Upload**	Users can upload images along with their found items to aid identification.
10	**Item Description**	Users can write detailed descriptions of items to help others identify them easily.

4.2. User Interaction Features

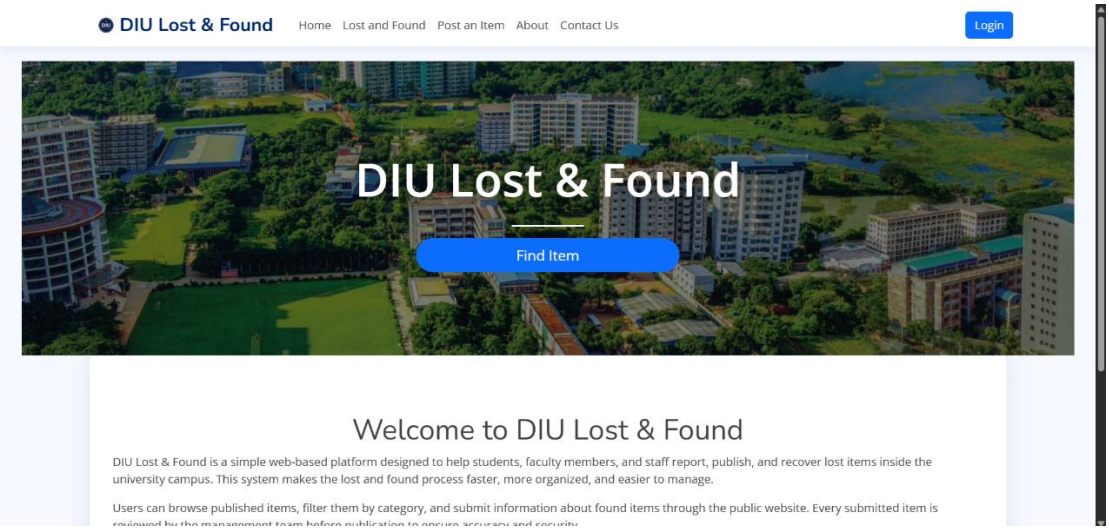
No.	Feature	Description
11	**Filter by Date**	Users can filter items based on the date they were found.
12	**Pagination**	Limits the number of items shown per page, making the interface more navigable.
13	**Item Claiming**	Allows users to claim items they believe belong to them.
14	**Claim Status**	Users can view the current status of claimed or unclaimed items.
15	**Admin Notifications**	Admins are notified of new posts, claims, or user inquiries.
16	**Lost and Found Item Submission**	Users can submit lost or found items through a simple form.
17	**Message Management**	Admins can manage user inquiries and respond to questions about items.
18	**User Activity Tracking**	Tracks user actions such as posting, claiming, and messaging for transparency.
19	**Sort Items by Condition**	Users and admins can sort items based on their condition (e.g., new, used, damaged).
20	**Item Availability Status**	Allows users to check if an item is still available or has been claimed.

4.3. Advanced Features

No.	Feature	Description
21	**Searchable Database**	The database is searchable, allowing for quick look-ups of items or users.
22	**User Management (Admin Panel)**	Admins can manage user accounts and permissions through the backend.
23	**Database Backup**	Admins can back up the system's database to prevent data loss.
24	**Item Sorting**	Users and admins can sort items by different criteria such as date or popularity.
25	**Email Notification**	Users can receive email alerts about new items, claims, or updates.
26	**Multi-language Support**	The system supports multiple languages, enabling access for a broader audience.
27	**User Activity Log**	Tracks user actions, such as item postings and claims, to ensure accountability.

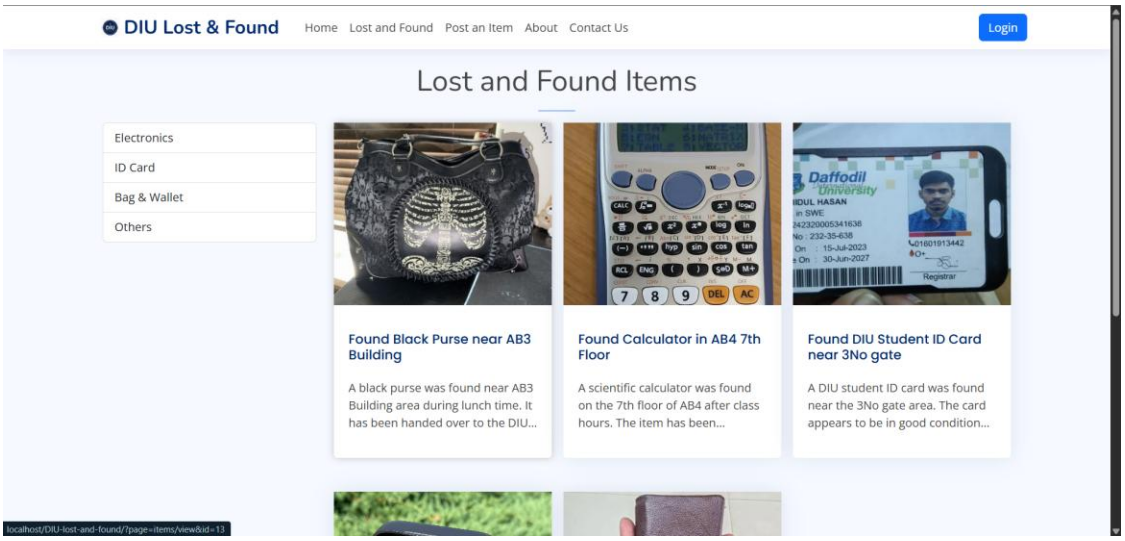
5. Screenshots and Functional Demonstrations

5.1. Homepage Overview



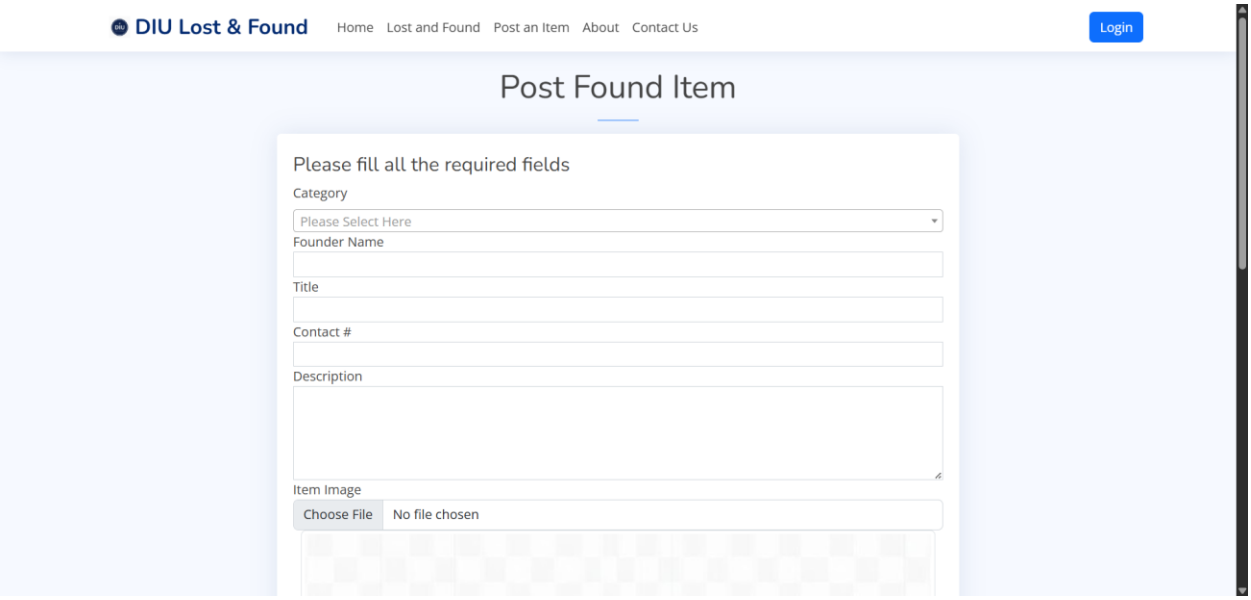
Description:
The homepage of the DIU Lost & Found system provides users with a clear, intuitive layout. It showcases the main features of the platform, such as the Find Item button for easy access to browsing lost and found items. The homepage also includes a welcoming description of the platform's functionalities, making it easy for users to understand how to report or recover lost items on the campus.

5.2. Lost and Found Items Page



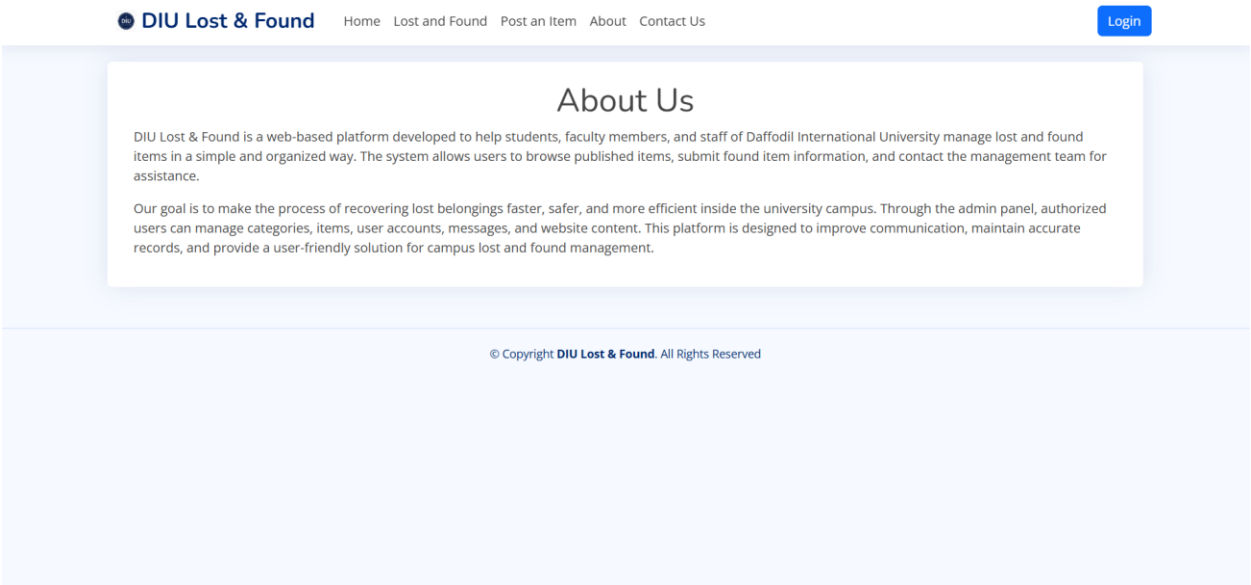
Description:
This page displays the list of found items posted by users. Items are organized by category, such as Electronics, Bags & Wallets, and ID Cards. Each item listing provides detailed information, including a description, location, and contact details for the item's finder, helping users easily identify and retrieve their lost belongings.

5.3. Post Found Item Page



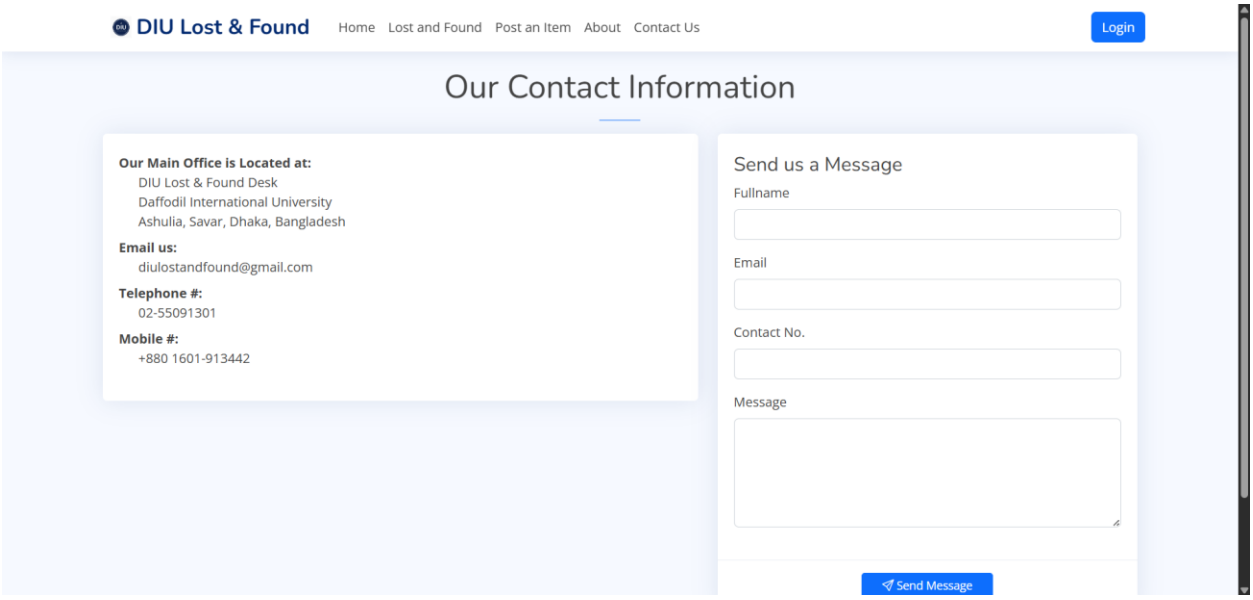
Description:
This page allows users to post a found item by filling in necessary details such as the category, founder's name, contact information, item description, and image upload. It ensures that the found item is properly documented, and users can share the item's relevant details with others on the platform. The form is designed to be user-friendly and helps in organizing the found items efficiently.

5.4. About Us Page



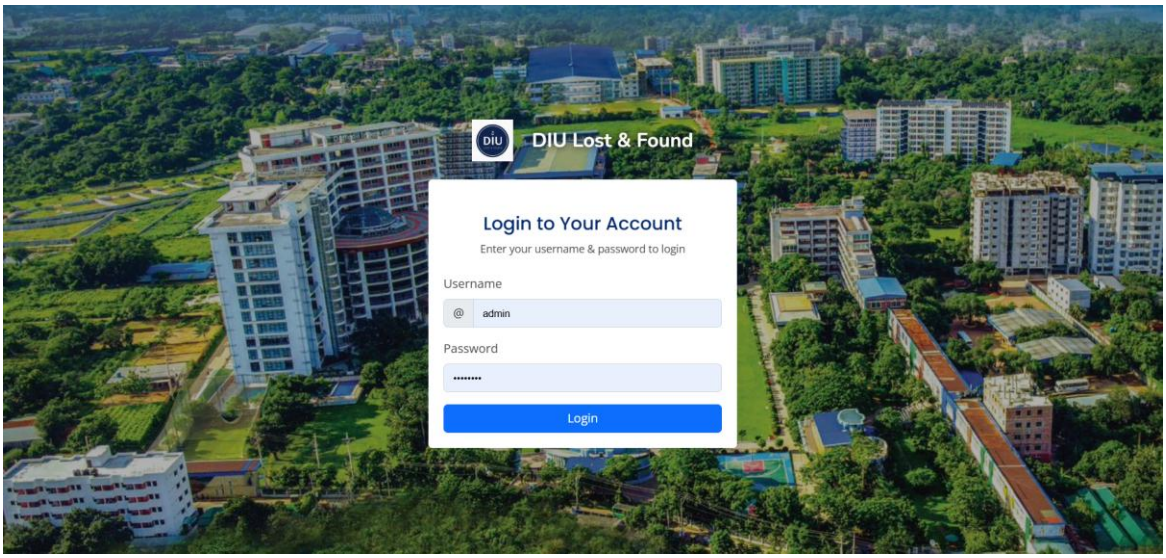
Description:
The About Us page of the DIU Lost & Found system provides an overview of the platform's purpose and objectives. It describes how the system helps students, faculty, and staff at Daffodil International University (DIU) by offering a centralized platform for reporting and recovering lost items. The page emphasizes the system's goal of improving communication, maintaining accurate records, and ensuring a user-friendly experience for all stakeholders involved.

5.5. Contact Us Page



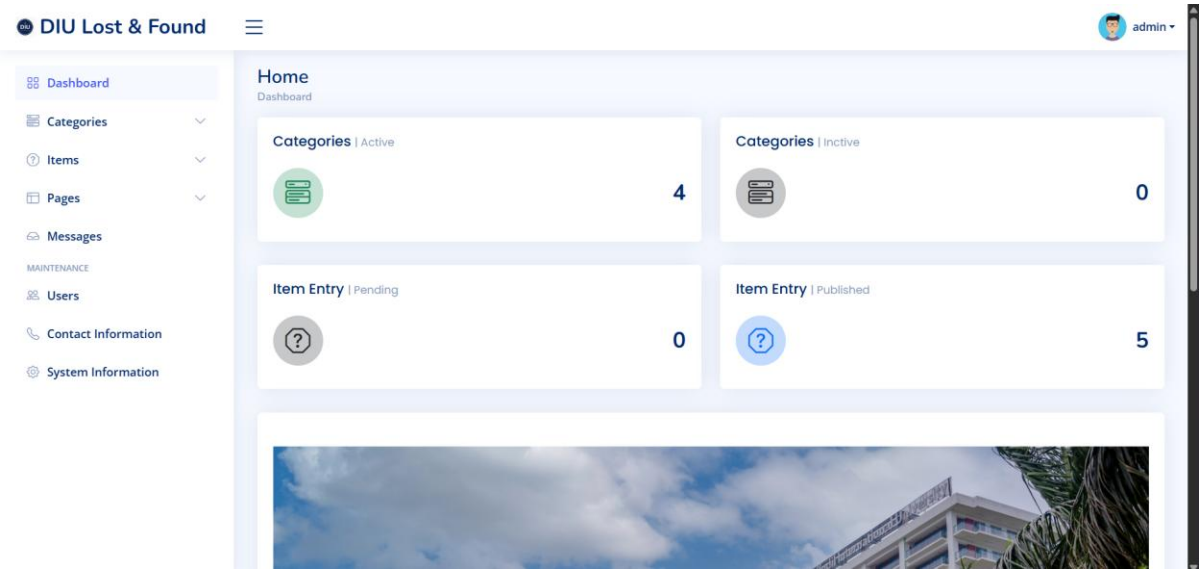
Description:
The Contact Us page provides essential information for users to get in touch with the DIU Lost & Found team. It includes the contact details such as the university's address, email, telephone, and mobile numbers. Additionally, users can directly send a message using the contact form, allowing them to inquire about lost or found items, or request further assistance regarding the system.

5.6. Admin Login Page



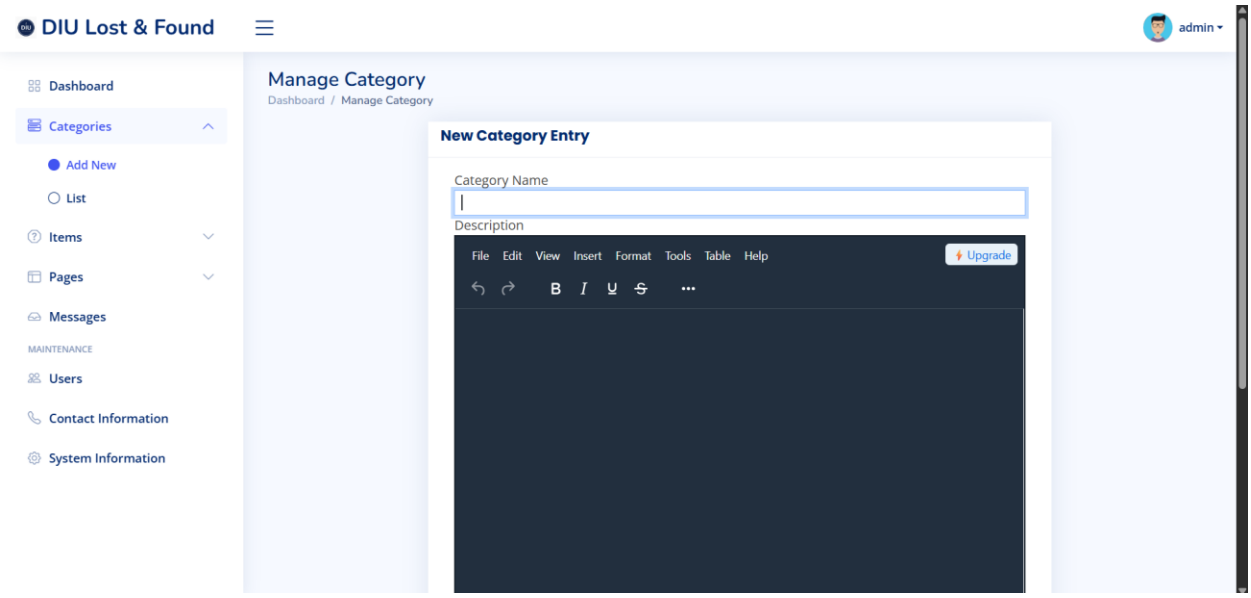
Description:
The Admin Login Page is exclusively for administrators to access the backend management features of the DIU Lost & Found system. Only authorized users, such as admins, can enter their username and password to log in and manage items, categories, and user interactions. This page is secure and ensures that all sensitive actions, like approving or rejecting found items and managing user data, are restricted to authorized personnel only.

5.7. Admin Dashboard



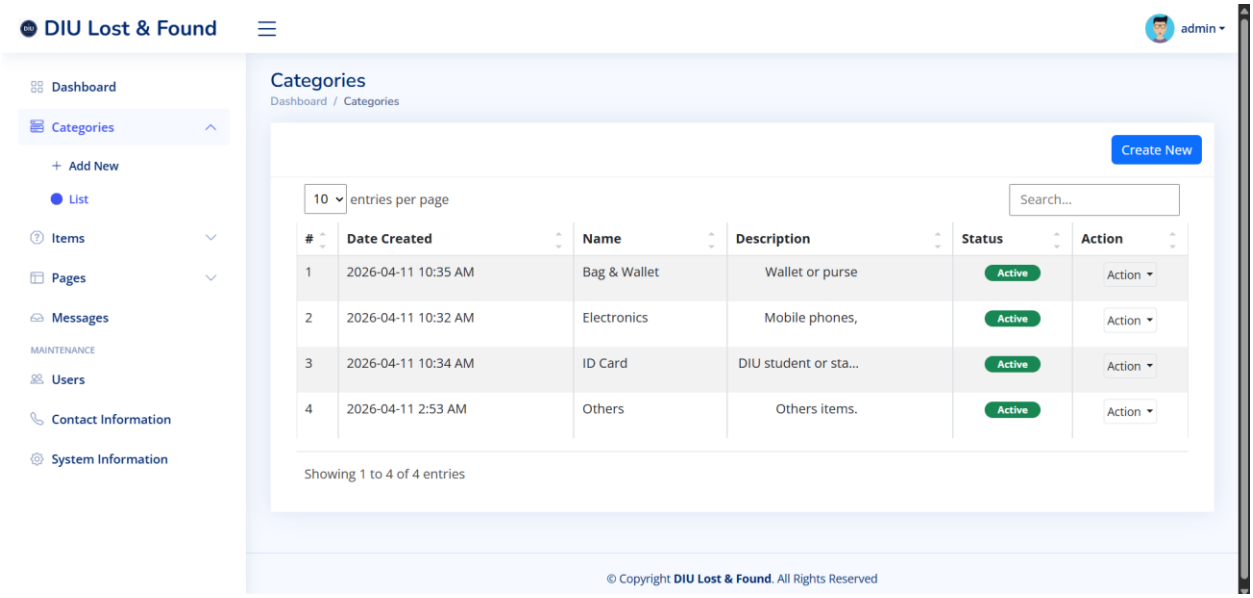
Description:
The Admin Dashboard offers an overview of the current system activity. It displays key metrics like the number of active and inactive categories, as well as the number of published and pending items. Admins can monitor the status of item entries, manage the items, and track categories easily. This feature provides admins with real-time insights into the system and simplifies the management of found and lost items within the university.

5.8. Manage Category Page



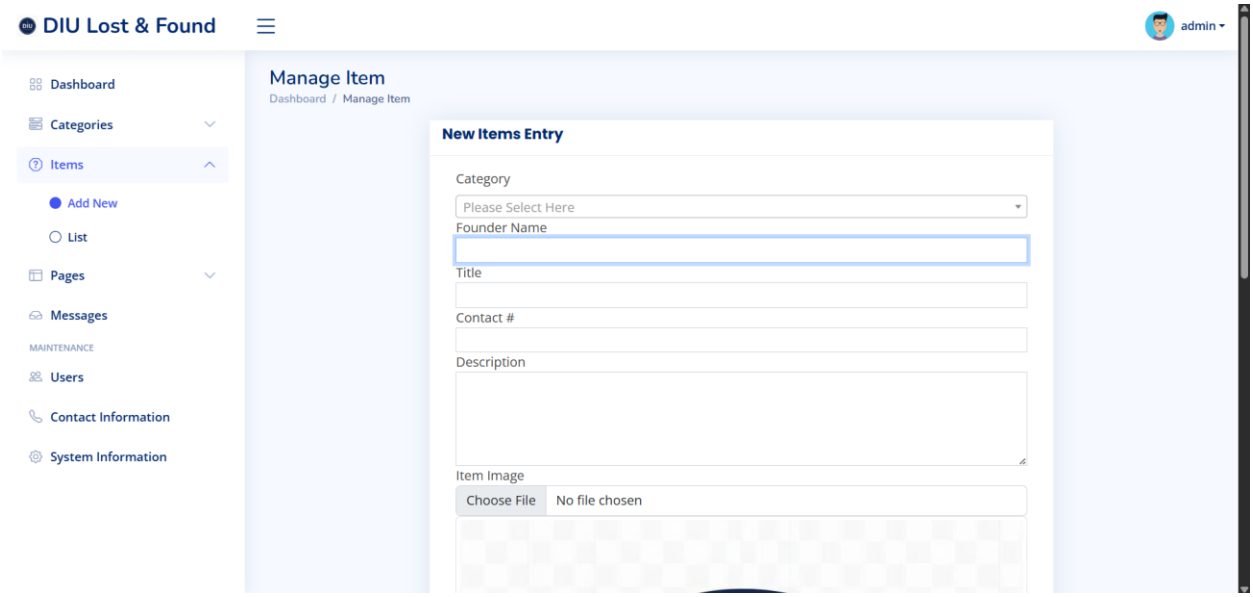
Description:
The Manage Category page allows administrators to create and manage the different categories for lost and found items. Admins can add a new category by entering its name and description, helping organize items effectively. The page provides a rich-text editor for the description, allowing admins to format it properly for better clarity. This feature ensures that items are classified appropriately and can be easily filtered based on their category.

5.9. Categories List Page



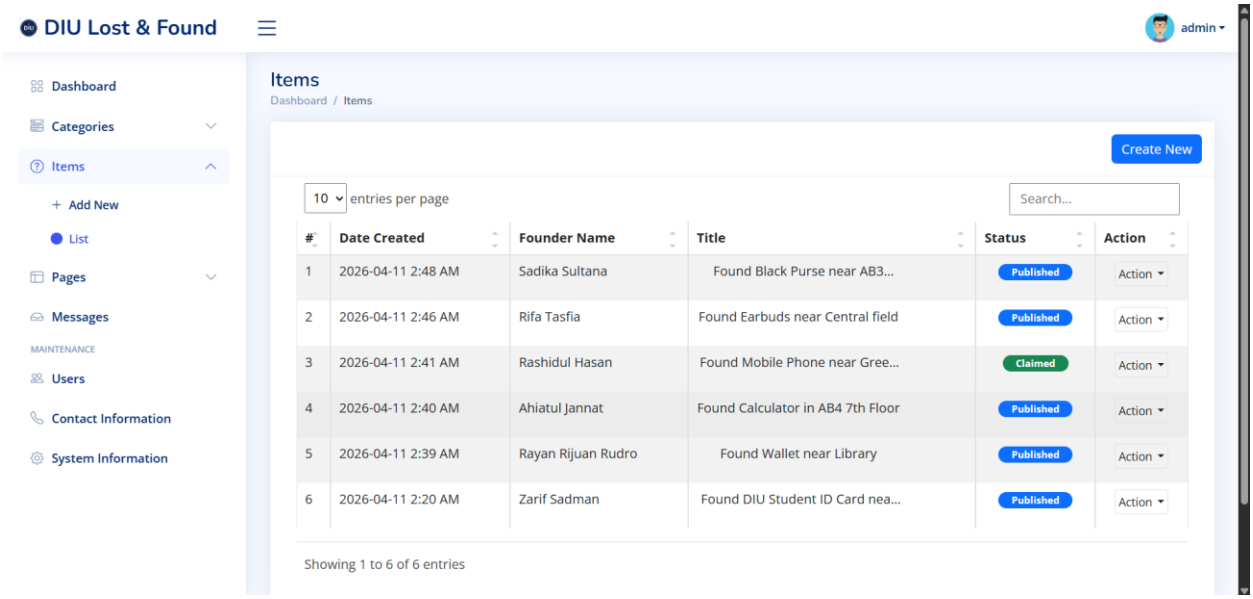
Description:
The Categories List page allows the admin to view and manage the different categories of lost and found items. This page shows the category name, description, and status (active or inactive) of each category. Admins can also edit or delete categories as needed. The system ensures proper categorization of items, making it easier for users to search and filter lost and found items.

5.10. Manage Item Page



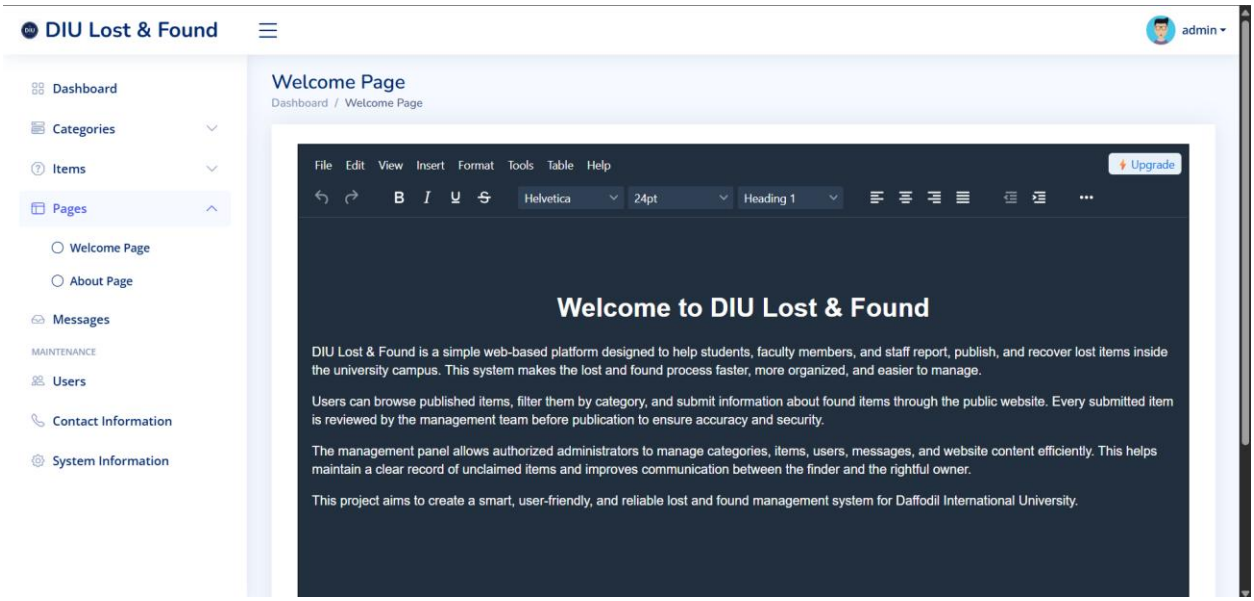
Description:
The Manage Item page allows administrators to add and manage lost and found items. Admins can select a category, provide the founder's name, give the item a title, and enter a description to provide further details. An image of the item can also be uploaded for better identification. This feature ensures that all lost and found items are recorded properly in the system for easy tracking and communication.

5.11. Items List



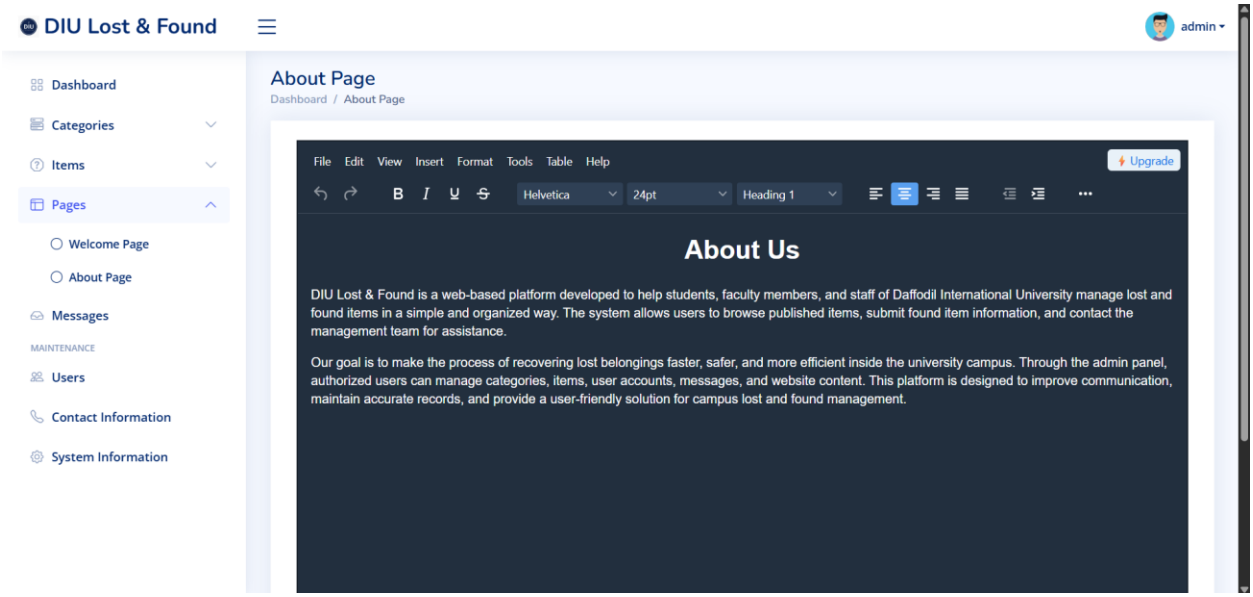
Description:
The Items List page in the admin dashboard displays all lost and found items that have been posted by users. Each entry shows the founder's name, the title of the item, and its status (e.g., Published, Claimed). Admins can easily manage items by editing, publishing, or marking items as claimed. This feature provides a clear overview of all the items in the system and helps admins track the progress of item claims.

5.12. Welcome Page



Description:
The Welcome Page serves as the first point of contact for users visiting the DIU Lost & Found system. It offers a brief introduction to the platform, highlighting its purpose of helping students, faculty, and staff with the reporting, publishing, and recovery of lost items within the university campus. The page emphasizes the system’s efficiency in making the lost and found process faster and more organized. Additionally, users are encouraged to browse published items and submit found items via the provided interface. The Welcome Page is the key starting point for all user interactions with the system.

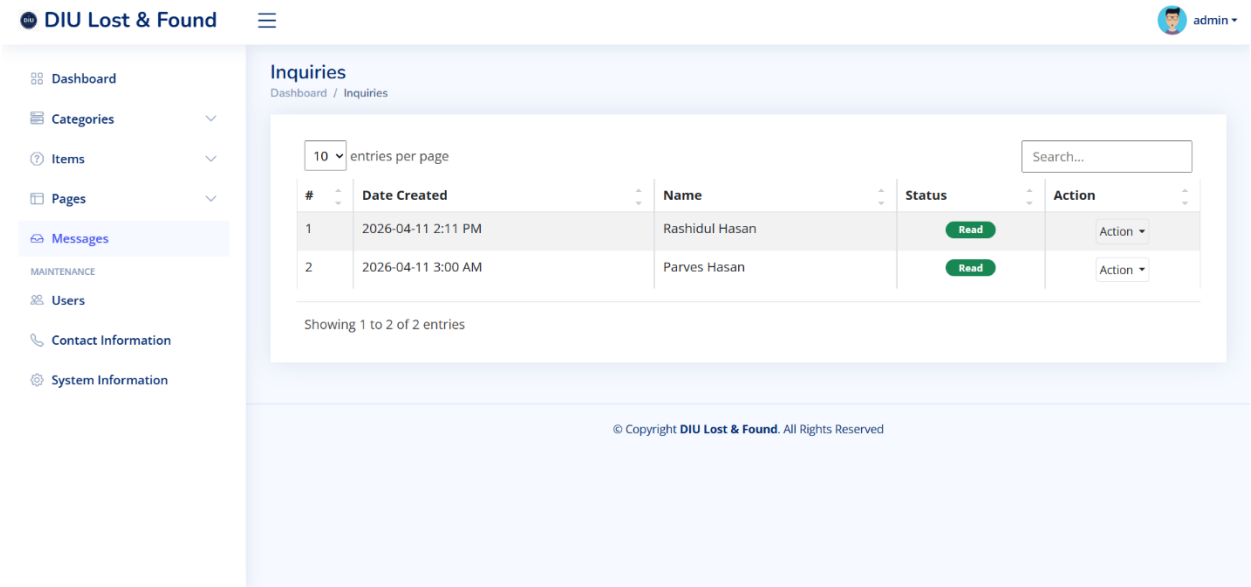
5.13. About Us Page



Description:

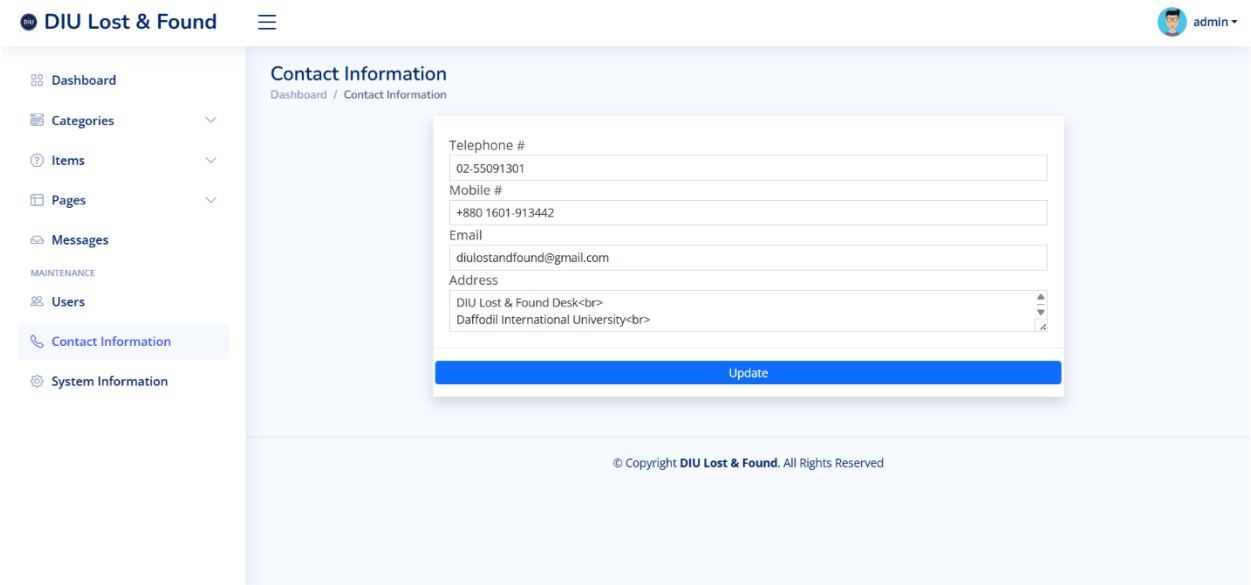
The About Us Page provides a detailed overview of the DIU Lost & Found platform's purpose and its goal within the university. It highlights the core mission of the system, which is to help students, faculty, and staff efficiently manage lost and found items within the campus. The page also outlines the platform's ability to allow authorized users (such as admins) to manage categories, items, user accounts, and messages. The About Us Page helps users understand the broader impact of the platform in terms of its functionality, as well as the ease and security with which lost items can be recovered.

5.14. Messages - Inquiries



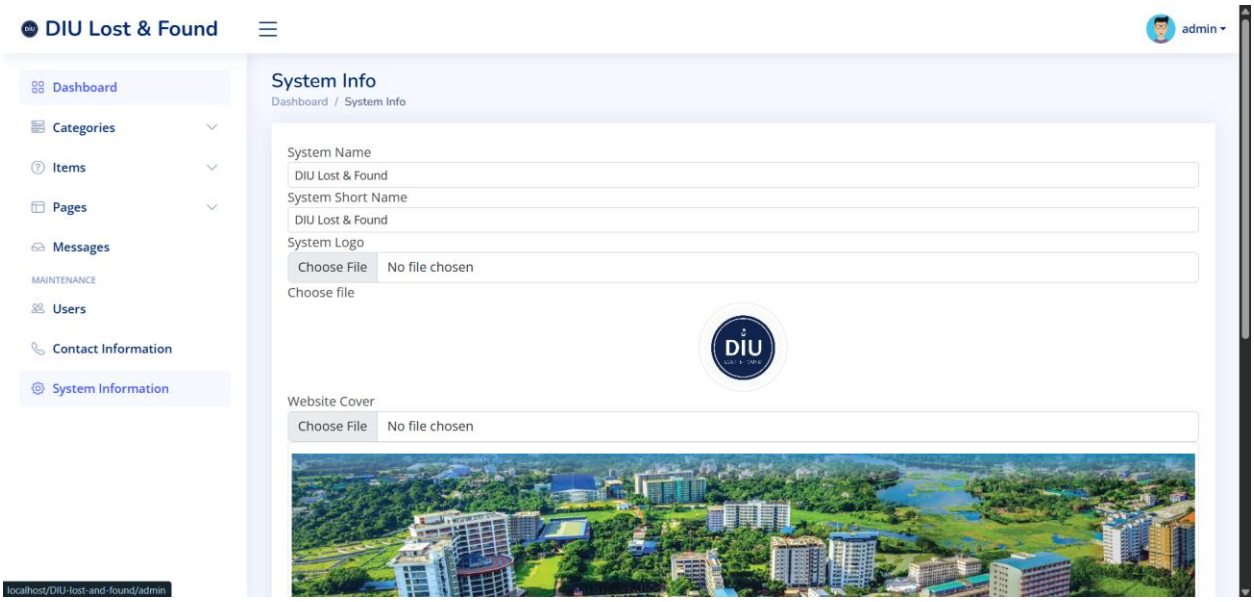
Description:
The Messages section allows the administrator to manage and review inquiries from users regarding lost and found items. It provides a table view of all received messages, including the date they were created, the name of the person who made the inquiry, and the status of each message. The Read status indicates whether the admin has reviewed the message. This feature ensures that all user inquiries are tracked, and appropriate actions can be taken for follow-up.

5.15. Contact Information Management



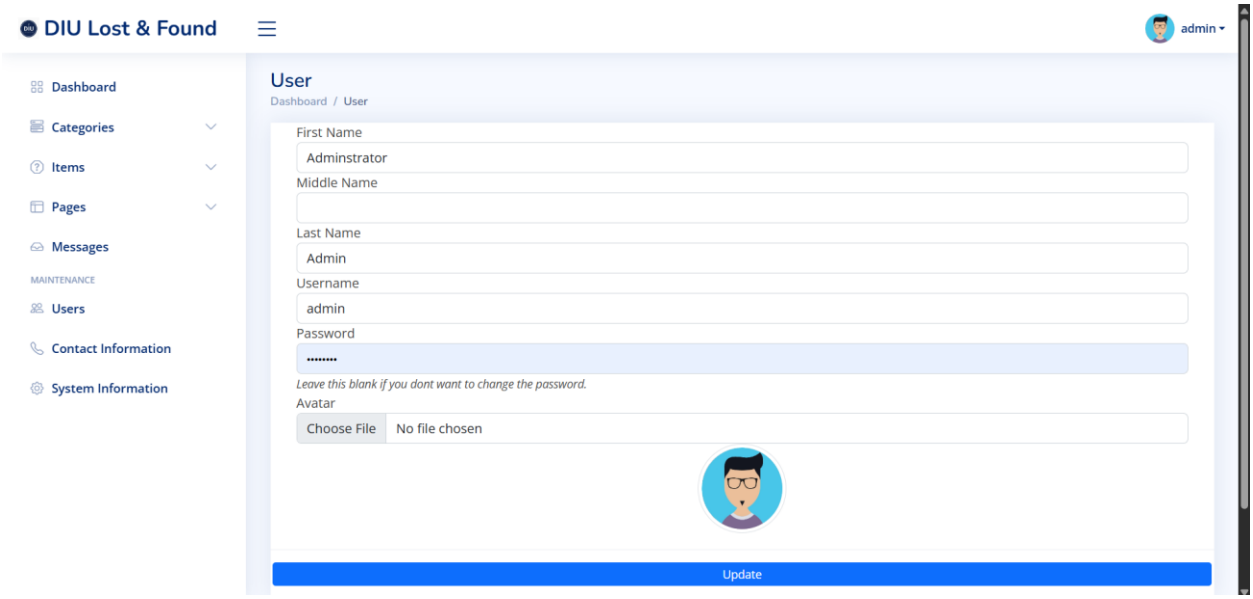
Description:
The Contact Information page allows administrators to manage and update the contact details for the DIU Lost & Found system. This includes important information such as the telephone number, mobile number, email address, and the physical address of the Lost & Found Desk at Daffodil International University. The admin can update these details to ensure that users can easily contact the right department for assistance. The feature enhances communication and ensures that the contact details are accurate and accessible at all times.

5.16. System Information Management



Description:
The System Information page allows administrators to update essential information related to the system, including the System Name, Short Name, Logo, and Website Cover. These settings help customize the DIU Lost & Found platform, allowing for branding and personalization of the user interface. The logo and cover image help maintain consistency with the university's identity, making the platform recognizable and visually appealing for users. This page is crucial for maintaining accurate branding and ensuring that the platform is properly identified as part of the university system.

5.17. Admin User Information Management



Description:

The Admin User Information Management page is an essential tool for system administrators. It allows the admin to update their personal details such as first name, last name, username, and avatar. This page is exclusive to the admin role, and it enables modifications to the admin's login credentials (username and password). Ensuring that the admin's information is accurate and up to date is crucial for maintaining secure access to the system and managing lost and found items effectively. After making necessary updates, the admin can save the changes by clicking the "Update" button.

6. System Workflow

6.1. User Interaction Flow

1. **Homepage**: Users visit the homepage to browse found items or post a new item. They can filter items by category or search for specific items.
2. **Post Found Item**: Users fill out a form to report a found item, including the item's category, title, description, contact number, and an image.
3. **Item Listing**: Once submitted, the item appears in the lost and found list for others to see. Pending items require admin approval.
4. **Messaging**: Users can send inquiries to admins if they need further information or wish to claim an item.

6.2. Admin Interaction Flow

1. **Admin Dashboard**: Admins log into a secure dashboard to manage items and categories. The dashboard provides an overview of pending and published items.
2. **Manage Items**: Admins can approve or delete found items. Items approved by the admin are published on the platform.
3. **Category & User Management**: Admins can add new categories, deactivate unused ones, and manage user roles.
4. **Responding to Messages**: Admins can respond to user inquiries and resolve issues related to found items.

7. Technical Analysis and Code Explanation

7.1. Core PHP Logic

- **index.php**: This is the main entry point where users interact with the system. It displays a list of found items and the search functionality.
- **found.php**: This page shows the detailed view of a found item, including its description and the ability to mark it as claimed.

- **post.php**: Handles the submission of found items, validating user input and saving item details in the database.
- **admin.php**: Admins use this page to manage the backend, approve items, and manage categories.

8. Evaluation

8.1. Testing

The system has undergone extensive testing, focusing on:

- **Functionality**: Ensuring that all forms (item posting, messaging, etc.) work as expected.
- **Usability**: Ensuring the interface is intuitive and user-friendly for both regular users and admins.
- **Security**: Verifying that user data is securely handled, with no vulnerabilities like SQL injection.

8.2. Performance

- **Speed**: The application loads promptly even with large datasets due to efficient database queries.
- **Scalability**: The system can handle an increasing number of items and users, thanks to the scalable MySQL database.

9. Conclusion

The **DIU Lost & Found** system is an efficient and scalable solution for managing lost items on the university campus. It provides a streamlined process for reporting, tracking, and recovering lost items. The system is secure, with proper user management and data integrity measures in place. It also offers a responsive user interface, ensuring a positive user experience.

9.1. Future Enhancements

- **Real-Time Notifications**: Implement notifications to alert users when their lost items are found.
- **Mobile App**: Develop a mobile app to make the system more accessible to users.

- **AI Integration**: Use machine learning for image-based recognition to automatically categorize items.

10.References

GitHub: https://github.com/rashidulhasan638/DIU-lost-and-found

Public site: http://localhost/DIU-lost-and-found/

Admin pane: http://localhost/DIU-lost-and-found/admin

phpMyAdmin: http://localhost/phpmyadmin